VICTORY 2027

FORCE WORK POWER & ENERGY

LECTURE-07 PHYSICS

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Topics To Be Covered

1. REVISION
2. DIFFERENT FORMS OF ENERGY
3. PRINCIPLE OF CONSERVATION OF ENERGY
4. QUESTIONS


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Physics Wallah

$$W_{net} = K_f - K_i$$

Work - Energy Theorem.

$$W = Fd\cos\theta$$

$\theta = 180^\circ$



$$W = -Fd$$

QUESTIONS

Imp A bullet of mass 5g travels with a speed of 500ms^{-1}. If it penetrates a fixed target which offers a constant resistive force of 1000N to the motion of the bullet, find:

(a) the initial kinetic energy of the bullet,

(b) the distance through which the bullet has penetrated before coming to rest, and

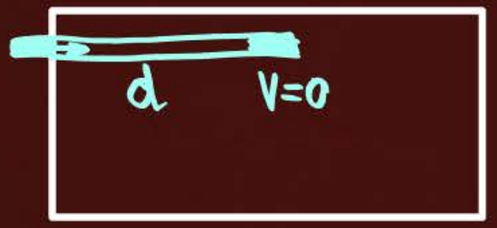
(c) the speed with which the bullet emerges out of the target if target is of thickness 0.5m.

(a) $K_i = \frac{1}{2}mu^2$

$= \frac{1}{2} \times \left(\frac{5}{1000}\right)(500)^2$

$= 625\ \text{J}$

(b)

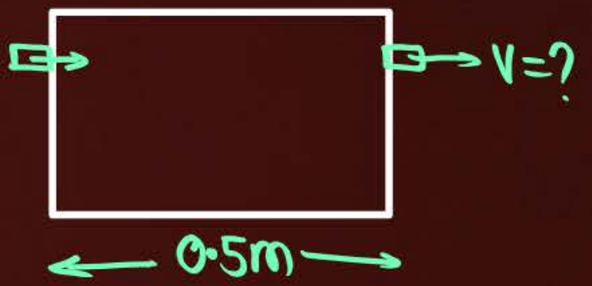


$W = K_f - K_i$

$-1000\,d = 0 - 625$

$\boxed{d = 0.625\ \text{m}}$

(c)



$W = K_f - K_i$

$-1000 \times 0.5 = \frac{1}{2}mv^2 - 625$

$-500 = \frac{1}{2}\left(\frac{5}{1000}\right)v^2 - 625$

$125 = \frac{5}{2000}v^2 \Rightarrow v = \sqrt{50000} = 223.6\ \text{m/s}.$

QUESTIONS

1mb

A ball of mass 10g falls from a height of 5m. It rebounds from the ground to a height of 4m. Find:

(a) the initial potential energy of the ball, $U = mgh = \frac{10}{1000} \times 9.8 \times 5 = 0.49\,\text{J}$

(b) the kinetic energy of the ball just before striking the ground, $K = U = 0.49\,\text{J}$

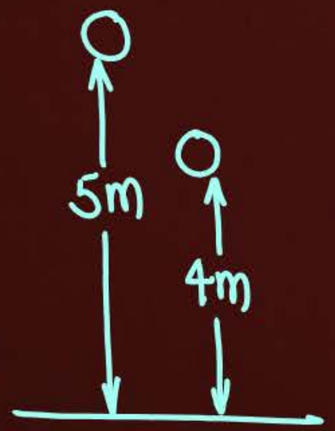
(c) the kinetic energy of the ball after striking the ground, and $K = U = mgh = \frac{10}{1000} \times 9.8 \times 4 = 0.392\,\text{J}$

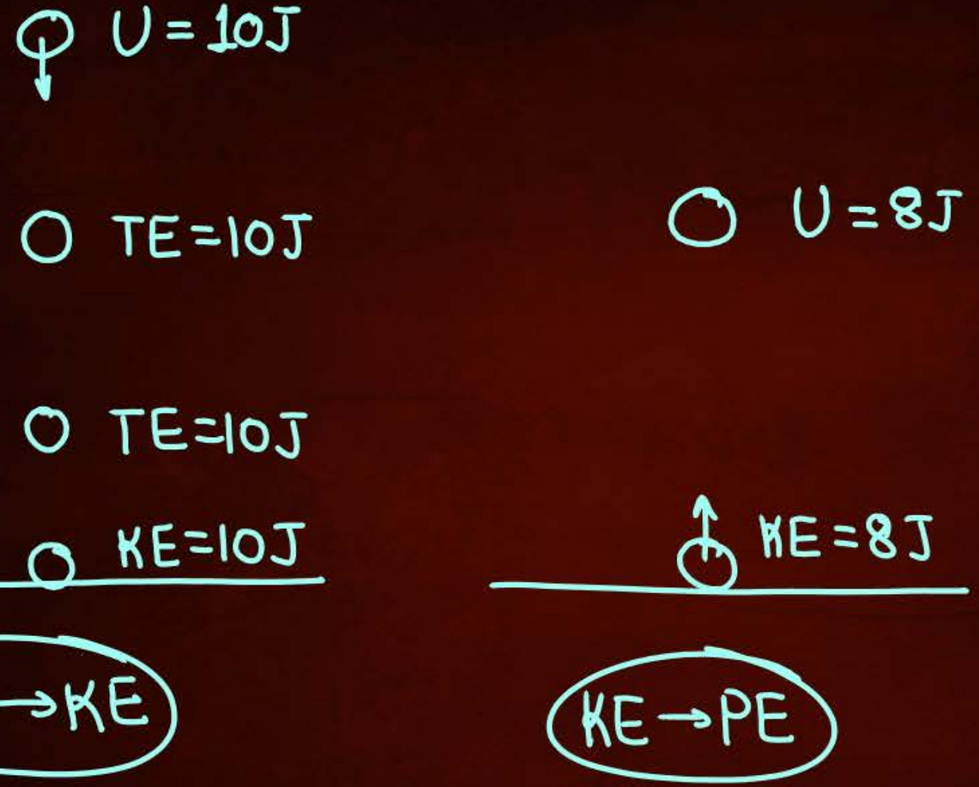
(d) the loss in kinetic energy on striking the ground.

Take $g = 9.8\text{ms}^{-2}$.

$\Delta K = 0.49 - 0.392$

$\boxed{\Delta K = 0.098\,\text{J}}$




U = 10J
TE = 10J
TE = 10J
KE = 10J
PE → KE
U = 8J
KE = 8J
KE → PE

QUESTIONS

A ball of mass 10g falls from a height of 5m. It rebounds from the ground to a height of 4m. Find:

(a) the initial potential energy of the ball,

(b) the kinetic energy of the ball just before striking the ground,

(c) the kinetic energy of the ball after striking the ground, and

(d) the loss in kinetic energy on striking the ground.

Take $g = 9.8\text{ms}^{-2}$.

Solution: Given: $m = 10\text{g} = \frac{10}{1000}\text{kg} = 0.01\text{kg}, h = 5\text{m}, g = 9.8\text{ms}^{-2}, h' = 4\text{m}.$

(a) Initial potential energy of the ball $= mgh = 0.01 \times 9.8 \times 5 = 0.49\text{J}$

(b) Kinetic energy of the ball just before striking the ground = Initial potential energy of the ball $= 0.49\text{J}$.

(c) Kinetic energy of the ball after striking the ground = Potential energy of ball at the highest point after rebound.

$$= mgh' = 0.01 \times 9.8 \times 4 = 0.392\text{J}$$

(d) Loss in kinetic energy on striking the ground = Initial kinetic energy - Final kinetic energy

$$= 0.49\text{J} - 0.392\text{J} = 0.098\text{J}$$

This energy appears in the form of heat energy and sound energy when the ball strikes the ground.

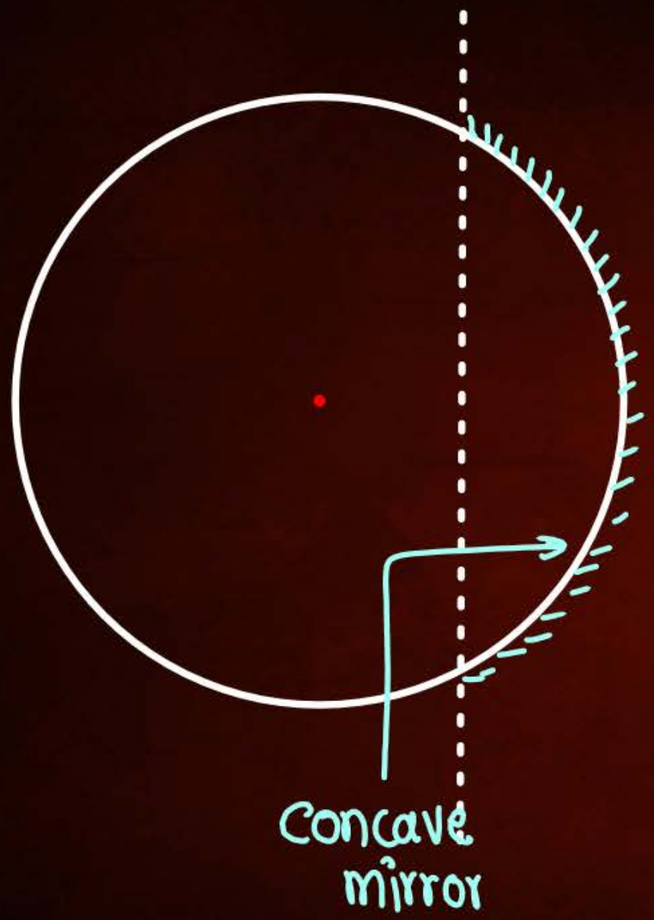
DIFFERENT FORMS OF ENERGY

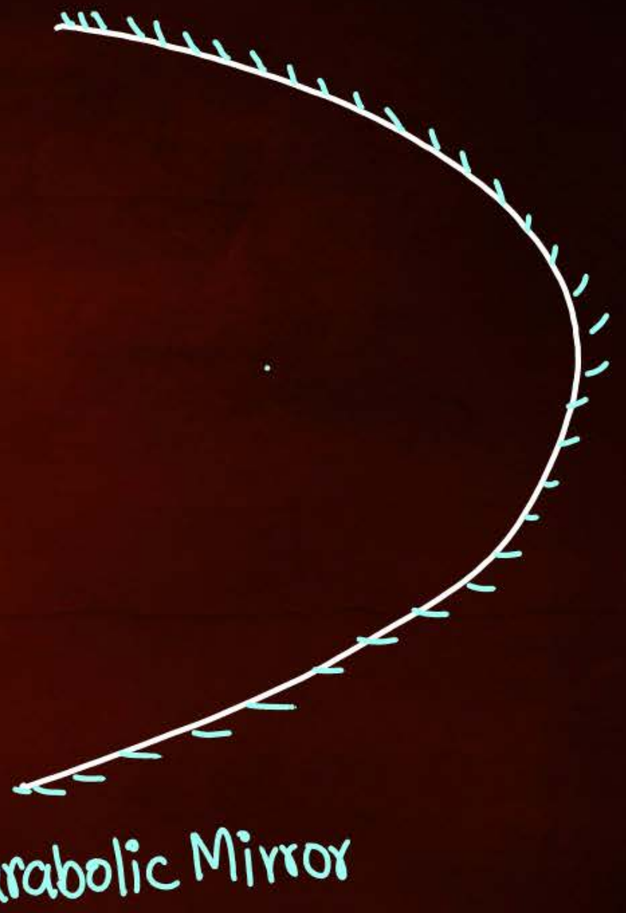
(1) **Solar energy:** The energy radiated out by sun is called the solar energy. Solar energy cannot be put to work directly, because it is too diffused (scattered by atmospheric particles) and it is not always available uniformly.

(i) A **solar panel** consists of black-painted metal surface which absorbs the sun's energy and heats water in its contact.

(ii) In a **solar furnace**, **parabolic mirror** is used to focus the sun rays on a small area so that a high temperature is achieved. Thus by keeping water at the focus of mirror, water can be boiled to produce steam and steam can then be used to drive the turbine of an electric generator so as to obtain the electrical energy.

(iii) The **solar cells** are used to convert the solar energy directly into the electrical energy. These cells are made from the thin slices of **semiconducting material**. However the efficiency of a solar cell to convert solar energy into the electrical energy is very low.


Concave
mirror


Parabolic Mirror

DIFFERENT FORMS OF ENERGY

(2) **Heat energy:** The energy released on burning coal, oil, wood or gas is heat energy. The steam obtained on heating water possesses heat energy and it has the capacity to do work. In a steam engine, heat energy of steam is used to obtain work (mechanical energy). We receive heat energy on earth directly from sun.

(3) **Light energy:** It is the form of energy in presence of which other objects are seen. The natural source of light energy is sun. The moon reflects the sun light at night. Many sources such as fire, burning candle, etc. also give light energy along with the heat energy.

DIFFERENT FORMS OF ENERGY

(4) **Chemical (or fuel) energy:** The energy possessed by the fossil fuels such as coal, petroleum and natural gas is called the chemical energy (or fuel energy). These fuels contain chemical energy stored in them. When fuels are burnt, the chemical energy changes into the heat and light energy.

The food that we eat also has the chemical energy in it. Plants convert the solar energy into the chemical energy of food during the process of photosynthesis. A battery has chemical energy stored in it and it provides the electrical energy when in use by a chemical reaction which takes place inside it.

(5) **Hydro energy:** The energy possesed by the fast moving water is called hydro energy. This energy is used to generate electricity in hydroelectric power stations. For this, water is stored in the dams built across the rivers high up in the hills and the water in the dams is allowed to fall down through pipes, the kinetic energy of the falling water drives the turbines of the generators which produces electrical energy.

DIFFERENT FORMS OF ENERGY

(6) **Electrical energy:** When two dry bodies are rubbed together, they get charged due to movement of free electrons from one body to the other body, so they possess electrical energy. An electric cell is a source of electrical energy due to movement of ions from one electrode to the other electrode within the cell as a result of a chemical reaction.

(7) **Nuclear energy:** The energy released due to loss in mass during the processes of nuclear fission and fusion, is called the nuclear (or atomic) energy. In both these processes the loss in mass gets converted into energy in accordance with the Einstein's mass-energy relation $\mathrm{E} = mc^2$. This energy is used in nuclear reactors for the constructive purpose to produce electrical energy and is used in atom bomb for the destructive purpose to produce heat and other forms of energy so as to destroy the enemy in war.

DIFFERENT FORMS OF ENERGY

(8) **Geo thermal energy:** The energy released in nuclear disintegrations in the interior of earth gets stored deep inside the earth and is called the geo thermal energy. This energy heats up the underground water to produce the natural steam. Sometimes, the natural steam may burst out from the surface of earth as hot springs. The electric power companies may drill wells into the earth to trap the natural steam which may be used to run the turbine of generators to produce electricity.

(9) **Wind energy:** The energy possessed by the fast moving air (i.e., storm) is called the wind energy. This energy is used in driving a wind mill. In past, for many years in rural areas, wind mills were used to pump out the underground water and to grind grains. Now a days, large giant wind mills are being used to drive the turbine of electric generators for producing electricity.

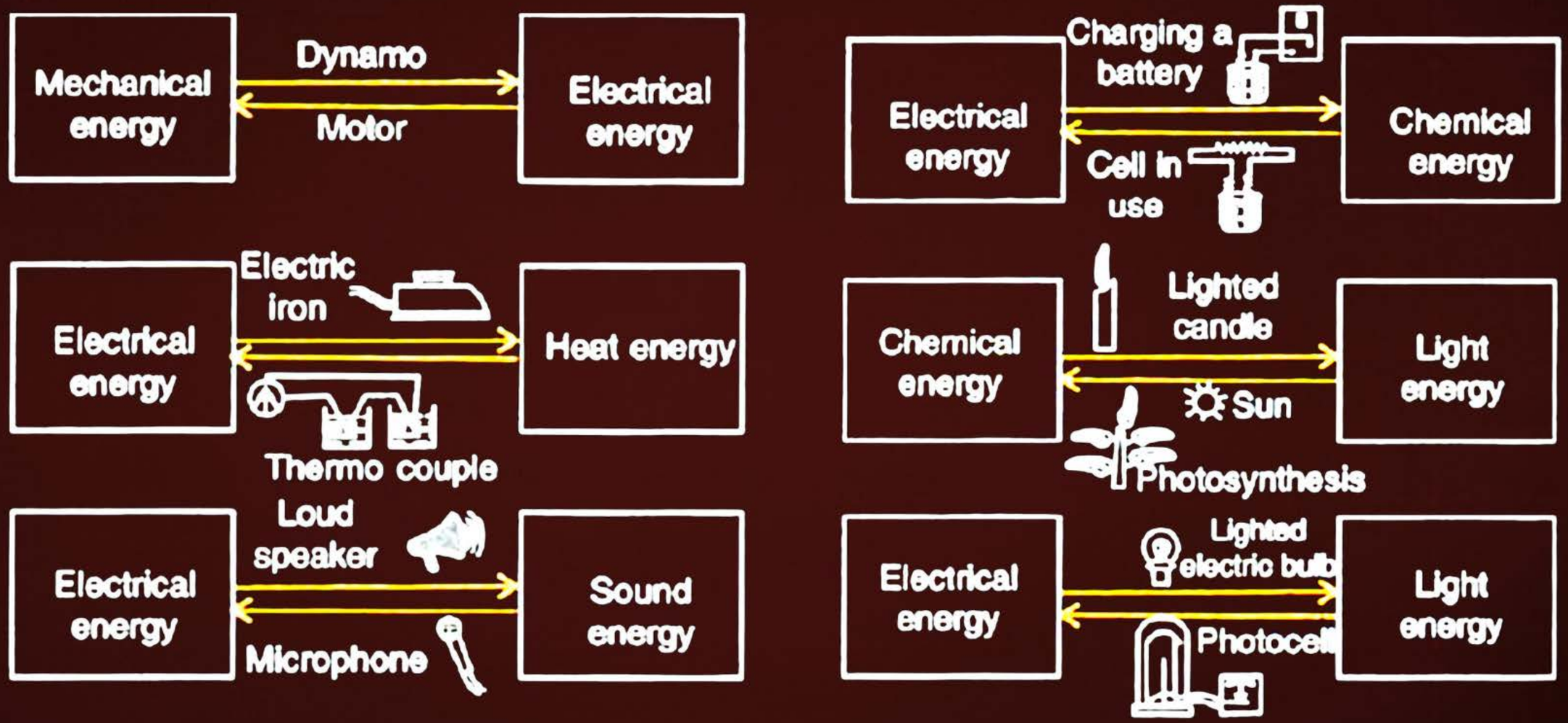
DIFFERENT FORMS OF ENERGY

10. **Sound energy:** A vibrating body possesses the sound energy. It is sensed by our ears. When the disturbance produced by a vibrating body in the atmospheric air layers reaches our ears and produces vibrations in the ear-membrane, sound is heard.

11. **Magnetic energy:** The energy possessed by a magnet due to which it can attract iron filings, is called the magnetic energy. An electromagnet has the magnetic energy.

12. **Mechanical energy:** The energy possessed by a body due to its state of rest or of motion, is called the mechanical energy. A body at a height, a moving body, a stretched bow, etc. have the mechanical energy.

CONVERSION OF ONE FORM OF ENERGY INTO THE OTHER FORM



QUESTIONS

A body at a height possesses:

1. kinetic energy
2. potential energy
3. solar energy
4. heat energy

QUESTIONS

In an electric cell while in use, the change in energy is from:

1. electrical to mechanical
2. electrical to chemical
3. chemical to mechanical
4. chemical to electrical.

QUESTIONS

During energy transformation some form of energy is given out without it being used anywhere, such an energy is called _____

1. backup energy
2. stored energy
3. wasteful energy
4. dissipated form of energy

QUESTIONS

In a washing machine the electrical energy is converted into _____

1. sound energy
2. vibrational energy
3. kinetic energy
4. mechanical kinetic rotational energy

QUESTIONS

The bullet weighing 100g is released from the barrel of an air gun with a velocity of 4ms^{-1}. Calculate the Potential energy of the spring.

1. 0.8 J
2. 8 J
3. 800 J
4. 0.08 J

$$U_s = K_B = \frac{1}{2}mv^2$$

$$= \frac{1}{2}\left(\frac{100}{1000}\right)(4)^2$$

$$= \frac{16}{20}$$

$$= \frac{4}{5}$$

PRINCIPLE OF CONSERVATION OF ENERGY

According to the principle of conservation of energy, energy can neither be created nor can it be destroyed. It only changes from one form to another.

- In universe, energy occurs in various forms. The sum of all forms of energy in the universe remains constant.
- When there is a transformation of energy from one form to another, the total energy always remains same i.e., it remains conserved.
- The principle of conservation of energy is one of the fundamental principles of nature.

HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.

QUESTIONS

What kind of energy is possessed in the situations when?

(i) A cocked up spring of air gun.

(ii) A stone lying on the top of roof.

(iii) A fish moving in water.

(iv) A horse running along a level road.

(v) Water stored in dams.

(vi) An electron spinning around nucleus.

(vii) A shooting arrow.

(viii) A stone in a stretched catapult.

Ans. (i) P.E. , (ii) P.E. , (iii) K.E. , (iv) K.E. , (v) P.E. , (vi) K.E. , (vii) K.E. , (viii) P.E.

CONVERSION OF ONE FORM OF ENERGY INTO THE OTHER FORM

(1) **Mechanical energy to electrical energy:** The water stored in the reservoir of a dam has the potential energy. When water falls, its potential energy decreases and kinetic energy increases. If the falling water is made to rotate a turbine near the bottom of the dam, the kinetic energy of water is transferred to the turbine in the form of rotational kinetic energy due to which it rotates. The turbine rotates the armature of the generator connected to it and thus the kinetic energy gets transformed into the electrical energy in the generator.

Thus an electric generator (or a dynamo) converts the mechanical energy into the electrical energy.

(2) **Electrical energy to mechanical energy:** In an electric motor, when an electric current is passed in a coil freely suspended (or pivoted) in a magnetic field, a torque acts on the coil due to which it rotates. The shaft attached to the coil also rotates with it. Thus the electrical energy changes into the mechanical energy.

The electric motor is used in many home appliances such as electric fan, washing machine, mixer, grinder, etc. It is also used to run the industrial machines.

(3) **Electrical energy to heat energy:** In electric appliances such as heater, oven, geyser, toaster, etc. electrical energy changes into the heat energy when a current passes through their resistance wire (or filament).

CONVERSION OF ONE FORM OF ENERGY INTO THE OTHER FORM

(4) **Heat energy to electrical energy:** In a thermocouple, when two junctions of two different metals are kept at different temperatures (one junction is kept hot, while the other cold), a current flows in the thermocouple. Thus a thermocouple changes the heat energy supplied at the hot junction into the electrical energy.

(5) **Electrical energy to sound energy:** A loudspeaker when in use, receives electrical energy in form of electrical signals from the microphone and changes it into the sound energy.

In an electric bell when an electric current is passed, the electrical energy changes into the sound energy.

(6) **Sound energy to electrical energy:** A microphone converts the sound energy into the electrical energy in form of varying electric signals.

(7) **Electrical energy to chemical energy:** While charging a battery, the electrical energy changes into the chemical energy of the cell.

(8) **Chemical energy to electrical energy:** From an electric cell when current is drawn, the chemical energy stored in it changes into the electrical energy.

(9) **Chemical energy to light energy:** When a candle burns, it gives light. Similarly in a kerosene lamp when the oil soaked in its wick burns, the chemical energy changes into the light energy.

CONVERSION OF ONE FORM OF ENERGY INTO THE OTHER FORM

(10) **Light energy to chemical energy:** The light energy from sun is absorbed by the green plants and they change it in form of the chemical energy during the process of photosynthesis.

(11) **Electrical energy to light energy:** When an electric bulb glows on passing an electric current through it, the electrical energy changes into the heat and light energies.

(12) **Light energy to electrical energy:** In a photoelectric cell, the light energy gets converied into the electrical energy.

In a solar cell, the light (or solar) energy changes into the electrical energy.

(13) **Heat energy to mechanical energy:** In a steam engine, the chemical energy of coal first changes to heat energy of steam and then heat energy of steam changes into the mechanical energy.

(14) **Chemical energy to heat energy:** When fuel such as wood, coal, bio-gas, etc. burns, the chemical energy changes into heat energy. In explosion of crackers, the chemical energy changes into the heat, light and sound energies. In lighting a candle or match stick, the chemical energy changes into the heat and light energies. In respiration, the chemical energy converts into the heat energy.

CONVERSION OF ONE FORM OF ENERGY INTO THE OTHER FORM

(15) **Chemical energy to mechanical energy:** In automobiles, while in motion, the chemical energy of petrol (or diesel) changes into the mechanical energy (or kinetic energy).

(16) **Electrical energy to magnetic energy:** While making an electromagnet, an electric current is passed in a coil wound around a soft iron bar which gets magnetised. Thus the electrical energy changes into the magnetic energy.

(17) **Mechanical energy to heat energy:** When water falls from a height, the potential energy stored in water at that height changes into the kinetic energy of water during the fall. On striking the ground (or bottom), a part of the kinetic energy of water changes into heat energy due to which the temperature of water rises.

The moving parts of a machine gets heated due to friction, thus a part of mechanical energy changes into heat energy.

CONVERSION OF ONE FORM OF ENERGY INTO THE OTHER FORM

Note:

(1) Whenever mechanical energy changes to other forms, it is always in the form of kinetic energy and not in the form of potential energy i.e., the stored potential energy first changes to kinetic energy and then kinetic energy changes to the other form.

(2) While transformation of energy from one form to the other desired form, the entire energy does not change into the desired form, but a part of it changes either to some other undesirable form (usually heat due to friction) or a part is lost to the surroundings due to radiation which is not useful. This conversion of energy to the undesirable (or non-useful) form is called the dissipation of energy. Since this part of energy is not available to us for any productive purpose, so we call this energy as the degraded from of energy.

Thank You